

Introduction to Quantum Approximate Optimization Algorithm (QAOA)

Presented by:

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Quantum Approximate Optimization Algorithm (QAOA)

- Quantum algorithm.
- Quantum computer is used to prepare and measure a quantum state.

A solution that is close to, but not exactly equal to the exact solution.

- Choosing the best possible option from a set of options.
E.g. maximizing profit, minimizing costs.

- **Types:**
 1. Continuous Optimization
 2. Discrete Optimization

- A recipe to solve a problem
 - **Types:**
 1. Classical Algo
 2. Quantum Algo (Classical problem via QA)

Agenda

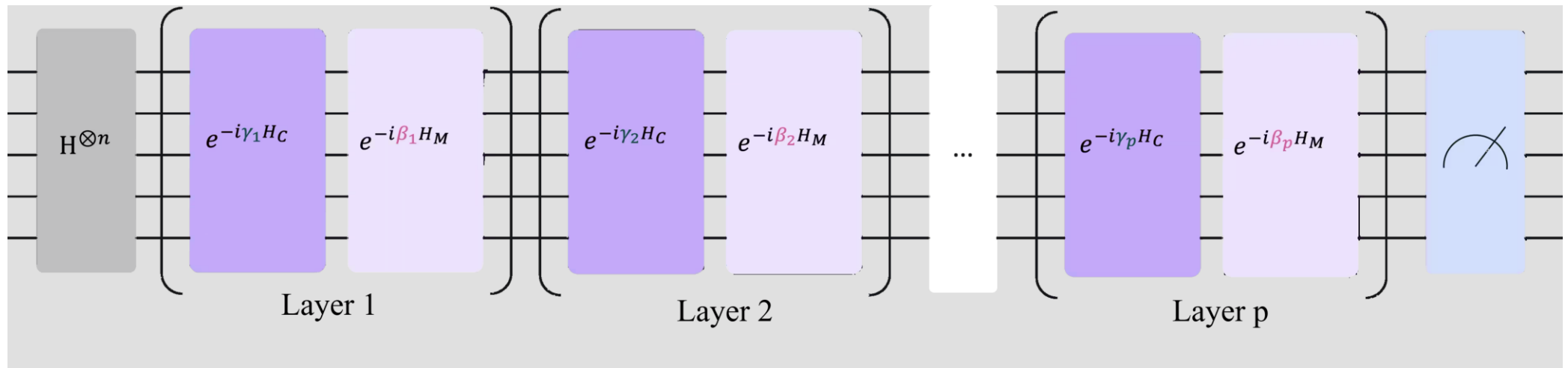
- › What is QAOA ?
- › QAOA as a discrete version of Quantum Adiabatic Algorithm(QAA)
- › Optimization problems: MaxCut and TSP
- › Solving MaxCut analytically (Exact solution)
- › QUBO matrix formalism

Agenda

- › Mathematical background of QAOA: lie theory and lie-trotterization
- › Derivation of unitary evolution operators $U_C(\gamma)$ and $U_B(\beta)$
- › Taylor expansion of these unitary operators and seeing corresponding quantum gates.
- › Finding net unitary $U(\beta, \gamma)$ and applying it onto some initial states like $|00\rangle$ and $|++\rangle$
- › Finding expectation values of cost Hamiltonian and finding optimized β and γ for $p = 1$ and 2 noded graph.
- › Composing simple QAOA circuits on Qiskit.

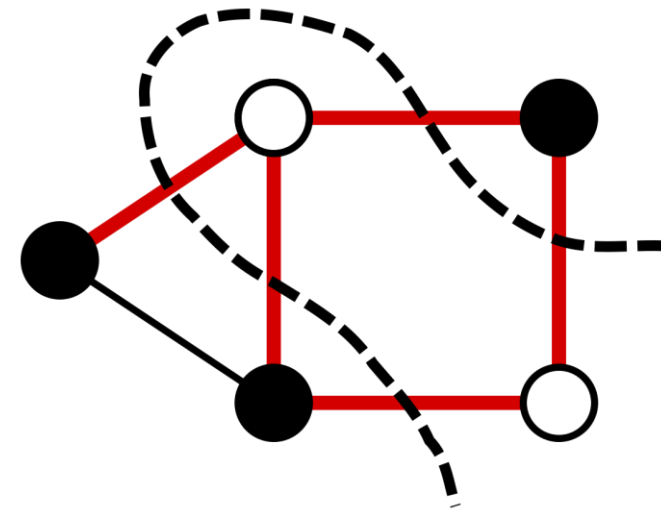
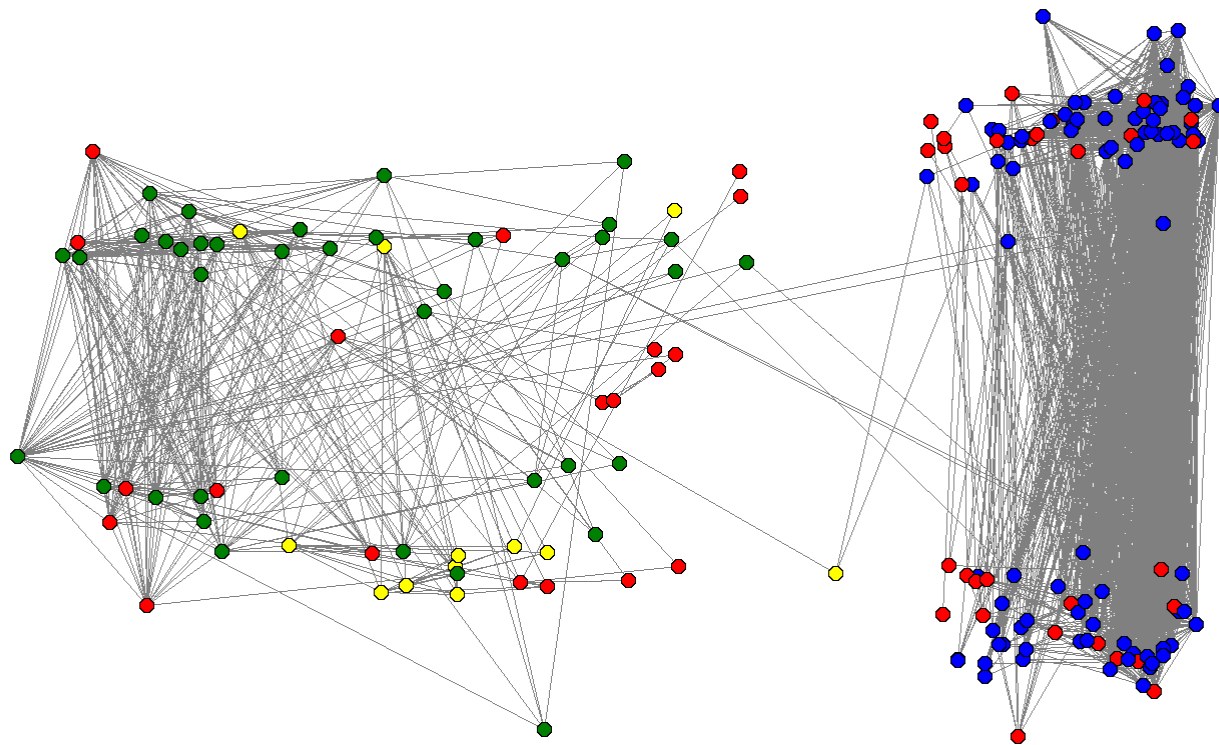
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Discretization



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MaxCut Problem



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TSP Problem



THE TRAVELLING SALESMAN PROBLEM

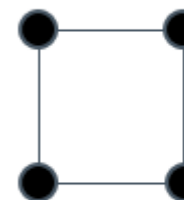
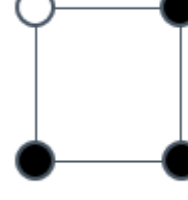
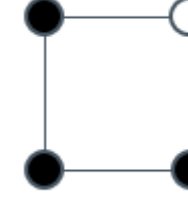
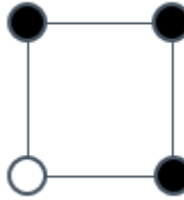
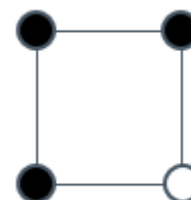
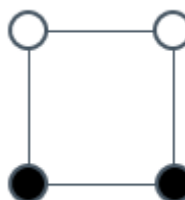
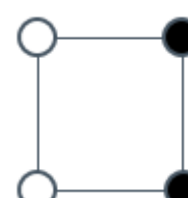
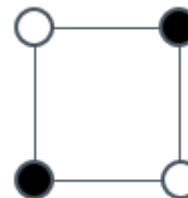
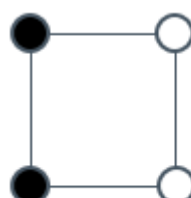
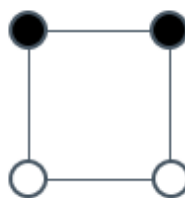
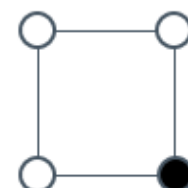
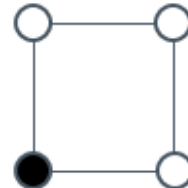
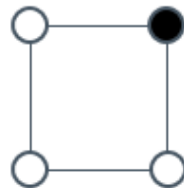
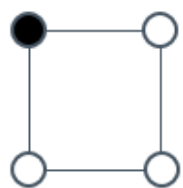
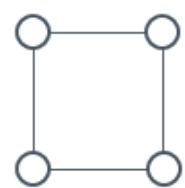
WHAT'S THE SHORTEST ROUTE TO VISIT ALL LOCATIONS AND RETURN?



ADDING MORE STOPS TAKES
LONGER AND LONGER AND LONGER TO FIGURE IT OUT

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MaxCut for four-noded graph



QUBO Formalism

$$\begin{aligned} f(x) &= x^T Q x = \sum_{i,j} Q_{ij} x_i x_j \\ &= \sum_{i=j} c_i x_i + 2 \sum_{i<j} Q_{ij} x_i x_j \end{aligned}$$

For Max-Cut problem

$$f(x) = x_i + x_j - 2x_i x_j$$

$$\begin{pmatrix} 2 & -1 & 0 & -1 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ -1 & 0 & -1 & 2 \end{pmatrix}$$

Qubo Matrix for four-noded graph

Ising Formalism

Classically, $f(x) = \sum_{i=j} c_i x_i + 2 \sum_{i<j} Q_{ij} x_i x_j$

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Spin Mapping, $x_i = \frac{1 - Z_i}{2}$

$$\hat{H}(z) = \text{constant} + \sum_{i=j} h_i \mathbf{Z}_i + \sum_{i<j} J_{ij} \mathbf{Z}_i \mathbf{Z}_j$$

For Max-Cut problem

$$\begin{aligned} h_i &= 0 \\ J_{ij} &= 1/2 \end{aligned} \quad C(Z) = \frac{1 - Z_i Z_j}{2}$$

Derivation of $U_c(\gamma)$ and $U_B(\beta)$

$$U_c(\gamma) = e^{-i\gamma H_c}$$

$$U_B(\beta) = e^{-i\beta H_B}$$

Time evolution unitary operator

$$U(t) = \exp\left(-\frac{i}{\hbar} \int_0^t H(s) ds\right) \approx \exp\left(-\frac{i}{\hbar} \sum_{j=0}^n H\left(j \frac{t}{n}\right) \frac{t}{n}\right)$$

❖ Using Lie-Trotter product formula,

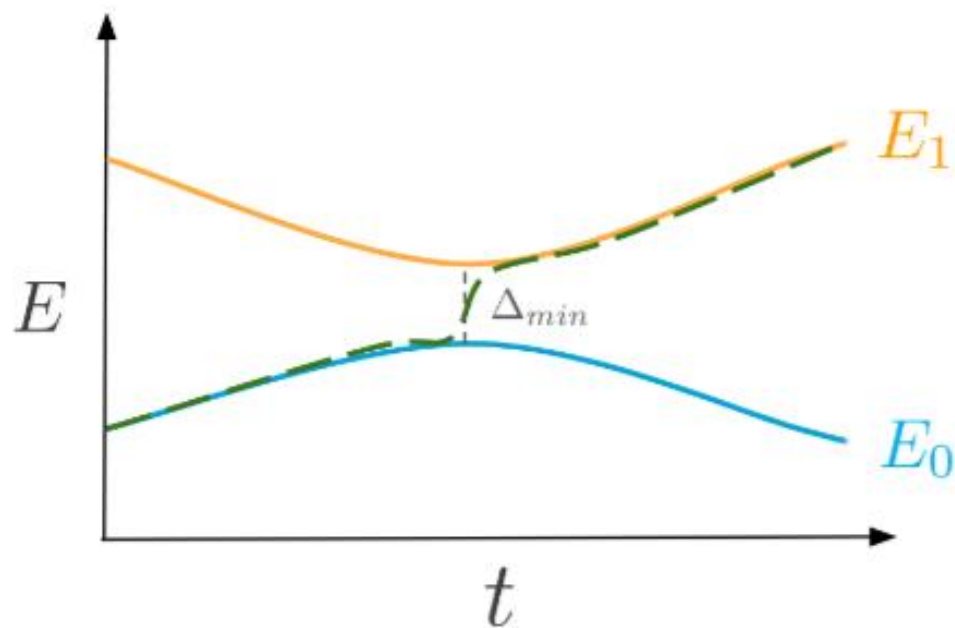
$$e^{A+B} \approx \lim_{n \rightarrow \infty} \left(e^{\frac{A}{n}} \cdot e^{\frac{B}{n}} \right)^n$$

$$U(t) = \lim_{n \rightarrow \infty} \prod_{j=1}^n \left[e^{-\frac{i}{\hbar} \left\{ H\left(j \frac{t}{n}\right) \frac{t}{n} \right\}} \right]$$

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Adiabatic Quantum Computing



Derivation (Continued)

❖ By Adiabatic Quantum Computation,

$$H(t) = \left(1 - \frac{t}{T}\right) \underbrace{(H_B)}_{\text{Easy Hamiltonian}} + \frac{t}{T} \underbrace{(H_C)}_{\text{Cost Hamiltonian}}$$

❖ By Trotter product formula

$$U(t) = \prod_{j=1}^p \underbrace{e^{-i\beta H_B}}_{U_B(\beta)} \underbrace{e^{-i\gamma H_C}}_{U_C(\gamma)}$$

Using these substitutions, at $t = T$ we get

$$\beta = \left(1 - \frac{j}{n}\right) \frac{t}{nm}$$

$$\gamma = \left(\frac{j}{n}\right) \frac{t}{nm}$$

Corresponding gates

By Taylor Expansion ,

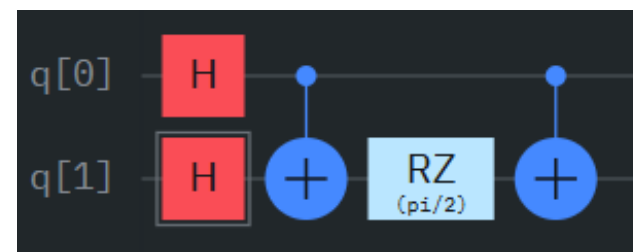
- $e^{-i\beta H_B} \approx \cos(\beta) I - i \sin(\beta) X$

- $e^{-i\gamma H_c} \approx \cos(\frac{\gamma}{2}) I + i \sin(-\frac{\gamma}{2}) Z_i Z_j \rightarrow \text{Controlled Phase Gate}$

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$U_B(\beta) = \begin{bmatrix} \cos \beta & -i \sin \beta \\ -i \sin \beta & \cos \beta \end{bmatrix}$$



$$R_Z(\theta) = \begin{bmatrix} e^{-i\theta/2} & 0 & 0 & 0 \\ 0 & e^{i\theta/2} & 0 & 0 \\ 0 & 0 & e^{i\theta/2} & 0 \\ 0 & 0 & 0 & e^{-i\theta/2} \end{bmatrix}$$

Let's see an example, how this algorithm works for MaxCut

$$p = 1,$$

$$|\Psi\rangle = U_B(\beta) U_c(\gamma) |\Psi_0\rangle$$

$$|\Psi_0\rangle = |++\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

$$U_B(\beta) \approx e^{-i\beta H_B} \approx \cos(\beta) I - i \sin(\beta) X$$

$$U_c(\gamma) \approx e^{-i\gamma H_c} \approx \cos\left(\frac{\gamma}{2}\right) I + i \sin\left(\frac{\gamma}{2}\right) Z_i Z_j$$

$$U_B(\beta) = \begin{bmatrix} \cos \beta & -i \sin \beta \\ -i \sin \beta & \cos \beta \end{bmatrix}$$

➤ Analytical Solution

- *Analytical derivation*

$$U_c(\gamma) \approx e^{-(i\frac{\gamma}{2})Z_i Z_j}$$

- *Derivation by gates*

$$U_c(\gamma) \sim CNOT RZ\left(-\frac{\gamma}{2}\right) CNOT$$

- Analytical derivation**

$$U_c(\gamma) \approx e^{-(i\frac{\gamma}{2})Z_i Z_j}$$

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$Z |0\rangle = +1 |0\rangle$$

$$Z |1\rangle = -1 |1\rangle$$

INPUT		OUTPUT		$Z_i Z_j$	$U_c(\gamma)$
i	j	Z_i	Z_j	-	-
$ 0\rangle$	$ 0\rangle$	$+ 0\rangle$	$+ 0\rangle$	$+ 00\rangle$	$e^{i\frac{\gamma}{2}} 00\rangle$
$ 0\rangle$	$ 1\rangle$	$+ 0\rangle$	$- 1\rangle$	$- 01\rangle$	$e^{-i\frac{\gamma}{2}} 01\rangle$
$ 1\rangle$	$ 0\rangle$	$- 1\rangle$	$+ 0\rangle$	$- 10\rangle$	$e^{-i\frac{\gamma}{2}} 10\rangle$
$ 1\rangle$	$ 1\rangle$	$- 1\rangle$	$- 1\rangle$	$ 11\rangle$	$e^{i\frac{\gamma}{2}} 11\rangle$

$$U_c(\gamma) |\Psi_0\rangle = \frac{1}{2} (e^{i\frac{\gamma}{2}} |00\rangle + e^{-i\frac{\gamma}{2}} |01\rangle + e^{-i\frac{\gamma}{2}} |10\rangle + e^{i\frac{\gamma}{2}} |11\rangle)$$

- Derivation by gates*

$$U_c(\gamma) \sim \text{CNOT} \text{RZ}\left(-\frac{\gamma}{2}\right) \text{CNOT}$$

$$U_c(\gamma) \sim \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} e^{i\gamma/2} & 0 & 0 & 0 \\ 0 & e^{-i\gamma/2} & 0 & 0 \\ 0 & 0 & e^{i\gamma/2} & 0 \\ 0 & 0 & 0 & e^{-i\gamma/2} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$U_c(\gamma) = \begin{bmatrix} e^{i\gamma/2} & 0 & 0 & 0 \\ 0 & e^{-i\gamma/2} & 0 & 0 \\ 0 & 0 & e^{-i\gamma/2} & 0 \\ 0 & 0 & 0 & e^{i\gamma/2} \end{bmatrix}$$

$$U_c(\gamma) |\Psi_0\rangle = \frac{1}{2} \begin{bmatrix} e^{i\gamma/2} & 0 & 0 & 0 \\ 0 & e^{-i\gamma/2} & 0 & 0 \\ 0 & 0 & e^{-i\gamma/2} & 0 \\ 0 & 0 & 0 & e^{i\gamma/2} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

$$U_c(\gamma) |\Psi_0\rangle = \frac{1}{2} (e^{i\frac{\gamma}{2}}|00\rangle + e^{-i\frac{\gamma}{2}}|01\rangle + e^{-i\frac{\gamma}{2}}|10\rangle + e^{i\frac{\gamma}{2}}|11\rangle)$$

$$|\Psi\rangle = U_B(\beta) U_c(\gamma) |\Psi_0\rangle$$

$$U_B(\beta) = \begin{bmatrix} \cos^2 \beta & -i \cos \beta \sin \beta & -i \cos \beta \sin \beta & -\sin^2 \beta \\ -i \cos \beta \sin \beta & \cos^2 \beta & -\sin^2 \beta & -i \cos \beta \sin \beta \\ -i \cos \beta \sin \beta & -\sin^2 \beta & \cos^2 \beta & -i \cos \beta \sin \beta \\ -\sin^2 \beta & -i \cos \beta \sin \beta & -i \cos \beta \sin \beta & \cos^2 \beta \end{bmatrix}$$

$$|\Psi\rangle = \begin{bmatrix} \cos^2 \beta & -i \cos \beta \sin \beta & -i \cos \beta \sin \beta & -\sin^2 \beta \\ -i \cos \beta \sin \beta & \cos^2 \beta & -\sin^2 \beta & -i \cos \beta \sin \beta \\ -i \cos \beta \sin \beta & -\sin^2 \beta & \cos^2 \beta & -i \cos \beta \sin \beta \\ -\sin^2 \beta & -i \cos \beta \sin \beta & -i \cos \beta \sin \beta & \cos^2 \beta \end{bmatrix} U_c(\gamma) |\Psi_0\rangle$$

for $|00\rangle$,

$$U_B(\beta) U_c(\gamma) |00\rangle = \frac{e^{i\frac{\gamma}{2}}}{2} \begin{bmatrix} \cos^2 \beta & -i \cos \beta \sin \beta & -i \cos \beta \sin \beta & -\sin^2 \beta \\ -i \cos \beta \sin \beta & \cos^2 \beta & -\sin^2 \beta & -i \cos \beta \sin \beta \\ -i \cos \beta \sin \beta & -\sin^2 \beta & \cos^2 \beta & -i \cos \beta \sin \beta \\ -\sin^2 \beta & -i \cos \beta \sin \beta & -i \cos \beta \sin \beta & \cos^2 \beta \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = e^{i\frac{\gamma}{2}} \begin{bmatrix} \cos^2 \beta \\ -i \cos \beta \sin \beta \\ -i \cos \beta \sin \beta \\ -\sin^2 \beta \end{bmatrix}$$

Similarly for $|01\rangle, |10\rangle, |11\rangle$

$$|\psi\rangle = \frac{1}{2} \begin{bmatrix} e^{i\gamma/2} \cos 2\beta - i e^{-i\gamma/2} \sin 2\beta \\ -i e^{i\gamma/2} \sin 2\beta + e^{-i\gamma/2} \cos 2\beta \\ -i e^{i\gamma/2} \sin 2\beta + e^{-i\gamma/2} \cos 2\beta \\ e^{i\gamma/2} \cos 2\beta - i e^{-i\gamma/2} \sin 2\beta \end{bmatrix}$$

After some simplifications ,

$$|\psi\rangle = \frac{1}{2} \begin{bmatrix} \cos(2\beta + \frac{\gamma}{2}) + i \sin(2\beta - \frac{\gamma}{2}) \\ \cos(2\beta - \frac{\gamma}{2}) - i \sin(2\beta + \frac{\gamma}{2}) \\ \cos(2\beta - \frac{\gamma}{2}) - i \sin(2\beta + \frac{\gamma}{2}) \\ \cos(2\beta + \frac{\gamma}{2}) + i \sin(2\beta - \frac{\gamma}{2}) \end{bmatrix}$$

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$$|\Psi\rangle = a_{00}|00\rangle + a_{01}|01\rangle + a_{10}|10\rangle + a_{11}|11\rangle,$$

$$\square a_{00} = a_{11} = \cos\left(2\beta + \frac{\gamma}{2}\right) + i \sin\left(2\beta - \frac{\gamma}{2}\right)$$

$$\square a_{01} = a_{10} = \cos\left(2\beta - \frac{\gamma}{2}\right) - i \sin\left(2\beta + \frac{\gamma}{2}\right)$$

?

$$\langle C \rangle = \langle \Psi | H | \Psi \rangle = -\frac{1}{8} [a_{00} \ a_{01} \ a_{10} \ a_{11}] \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{00} \\ a_{01} \\ a_{10} \\ a_{11} \end{bmatrix}$$

After some simplifications

$$\langle C \rangle = \frac{1}{2} (\sin 4\beta \sin \gamma)$$

$$(\beta, \gamma) = \left((2n+1)\frac{\pi}{8}, (2n+1)\frac{\pi}{2} \right)$$

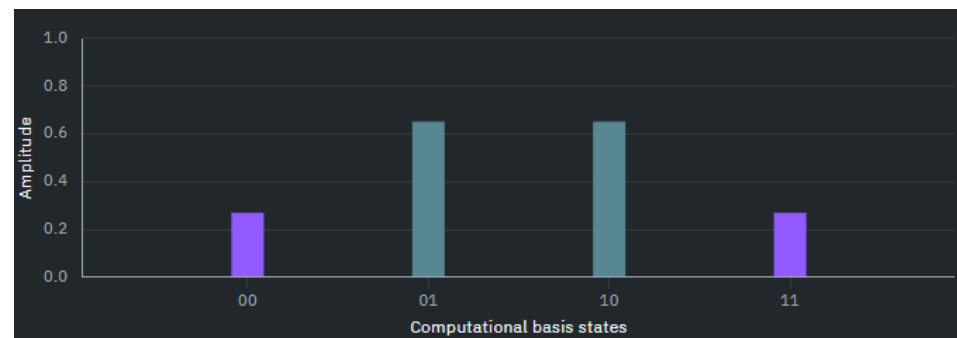
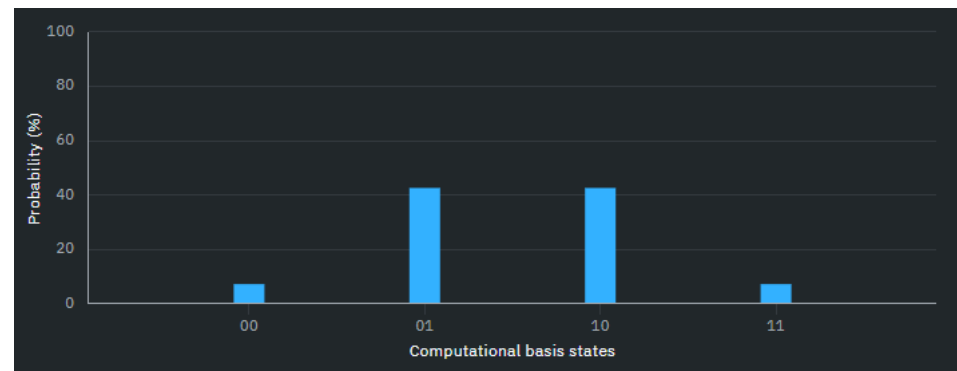
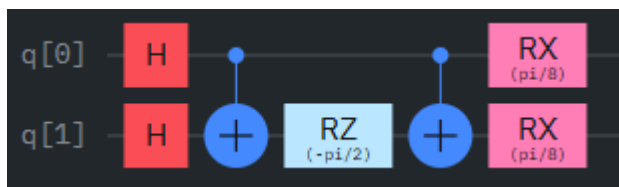
$$|\psi\rangle = \frac{1}{2} \begin{bmatrix} \cos\left(2\beta + \frac{\gamma}{2}\right) + i \sin\left(2\beta - \frac{\gamma}{2}\right) \\ \cos\left(2\beta - \frac{\gamma}{2}\right) - i \sin\left(2\beta + \frac{\gamma}{2}\right) \\ \cos\left(2\beta - \frac{\gamma}{2}\right) - i \sin\left(2\beta + \frac{\gamma}{2}\right) \\ \cos\left(2\beta + \frac{\gamma}{2}\right) + i \sin\left(2\beta - \frac{\gamma}{2}\right) \end{bmatrix}$$

Optimizing β and γ

$$\frac{\partial C}{\partial \beta} = 0 \quad \frac{\partial C}{\partial \gamma} = 0$$

QAOA circuit for 2 noded-graph

- for $p = 1$



Computational basis state	Probability
00	7.32233%
01	42.67767%
10	42.67767%
11	7.32233%

The background is a solid blue color with various white abstract elements. There are wavy lines, small triangles, and clusters of white dots scattered across the surface. The text 'THANK YOU' is centered in a white, hand-drawn, slightly irregular font.

THANK YOU